Train Once, Read Everywhere: Substrate Invariance of the Linearly Readable Structure in Frozen Language Models

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Abstract

We report the consolidated findings of the SRT (Semiotic-Reflexive Transformer) research program: a multi-year effort to treat frozen, production-scale language models not as black boxes to be fine-tuned but as *substrates* whose internal states carry structure that small, inspectable instruments can read. The program's individual results include a ~12M-parameter adapter that exposes per-token semiotic signals from a frozen 7B backbone with zero cross-entropy degradation; an activation verbalizer that recovers text from single hidden states up to a calibrated paraphrase ceiling; read-out ports spanning architectures from dense 3B to 94-layer 235B-parameter mixture-of-experts models; and a 22MB linear head that gives a frozen multimodal chat model image-to-text retrieval at the level of fully-trained 2018 dual encoders on the standard COCO benchmark.

The headline finding, established in the program's final arc, unifies these results: **the readable structure is a stable property of the model class, invariant along every axis a deployment can vary.** The cross-modal correspondence inside a frozen multimodal LLM is anisotropic-linear (a rotation cannot express it; a nonlinear model finds nothing beyond it at any data scale tested, and falls further behind as data grows). A head trained once on one host reads, with no retraining and at most a 42KB recalibration: a host ten times smaller (31B to 3B, no capability loss at matched data), the same host at 4-bit weight precision (a 0.011 loss in R@1 with unchanged head weights), and the same computation on entirely different silicon and kernels (CUDA/bf16 datacenter to Apple-Silicon/MLX-Q4, 97.0% head-space retrieval agreement over the full 5,000-caption pool against a 99.96% same-runtime reproducibility ceiling, and consumer-tier image-to-text retrieval within three R@1 points of the datacenter reference). Deployment tiers, from Raspberry-Pi-class edge devices to datacenter fleets, differ in latency and cost, never in capability. Train once, read everywhere.

We present the instrument stack, the measurement discipline that made the numbers trustworthy (anisotropy-corrected metrics, anchored reference frames, controls at every rung), the invariance evidence, and an honest ledger of the program's negative results, several of which are load-bearing for the headline claim.

Companion documents: the Stage-3 SRT-Adapter manuscript, the Stage-4 NLA manuscript, and the engineering guide, all in the program repository (https://github.com/space-bacon/SRT). Every number in this paper has a public artifact and a control behind it; the artifact index is Section 14.

1. Introduction

1.1 The claim

A frozen language model computes more than its output distribution. At every layer, for every token, it produces intermediate states, and the SRT program's founding bet was that these states carry *readable* structure: signals about meaning, divergence, reflexive awareness, and discourse position that a small instrument can extract without touching a single backbone weight.

The program's accumulated experiments support the bet, but the result that organizes all the others arrived last, and it is stronger than the bet itself. It is not merely that the structure is readable. It is that **what the instruments read does not depend on where or how the substrate runs.** The structure survives a ten-fold change in host scale. It survives quantization of the weights to four bits. It survives a change of hardware, numerical kernels, and quantization scheme simultaneously. And within a multimodal host it is shared across modalities, connected by a transformation so simple (a single linear map after per-modality centering) that its entire deployed form fits in 22 megabytes and can be audited by inspection.

We call this property **substrate invariance**, and we summarize it operationally: *train once, read everywhere.* An artifact calibrated against one instance of the model class reads every instance the deployment world produces, from an edge device performing a single prefill pass to a datacenter serving a fleet. The tiers differ in latency and cost. They do not differ in capability.

1.2 The program in six stages

1. **Stages 1–2 (theory and pretraining architecture).** The semiotic framework: signs, interpretants, attractor basins of meaning, and a proposed architecture with these structures built into the embedding layer (Lancaster 2025, SSRN 5987495; Lancaster 2026a, SSRN 6349978).
2. **Stage 3 (the adapter).** The realization that honest instrumentation beats reconstruction: freeze a strong existing model and bolt on ~12M trainable parameters that read what is already latent (`arxiv/paper.md`).
3. **Stage 4 (activation verbalization).** Reading a single hidden state back as text, with the anisotropy-corrected, anchored measurement frame that the rest of the program inherited (`paper_nla.md`).
4. **Stage 5 (scaling the read-out up).** Ports of the read-only adapter to gpt-oss-20b and Qwen3-235B-A22B, establishing architecture- and scale-generality upward.
5. **Stage 6 (crossing modalities: SRT-Sunstone).** The discovery that a text-trained read-out interprets images zero-shot on a frozen multimodal host, and the boundary studies that made the claim precise.
6. **The invariance arc (this paper's contribution).** The controlled ladder that identified the cross-modal gap as anisotropic-linear, benchmarked it, and then deliberately varied scale, precision, and hardware to establish that the readable structure is invariant.

1.3 How to read this paper

§2 describes the instruments. §3 describes the measurement discipline; we consider it a contribution in its own right, because nearly every number in this field is uninterpretable without it. §4 through §7 present the evidence, organized by invariance axis. §8 assembles the headline claim. §9 is the ledger of negative results. §10 gives the semiotic interpretation. §11–§13 situate, scope, and conclude.

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2. The instrument stack

All instruments share one design rule: **the backbone is frozen, byte-identical, and runs natively.** Instruments read hidden states from forward passes the host is already performing; a deployment gains capabilities without any risk of regression to the host's primary function.

2.1 The SRT-Adapter (Stage 3)

A ~12M-parameter module (about 0.17% of a 7B backbone) that taps the residual stream of frozen Qwen2.5-7B at three layers. Metapragmatic Attention Heads (MAH) read divergence; a GRU-based Reflexive Recurrent Module (RRM) integrates the divergence stream into a meta-state; a Bifurcation Estimation Network (BEN) emits per-token reflexivity $\hat{r}$ and a regime label; a community head discovers discourse-trajectory structure without labels; and an optional FiLM correction ($\mathtt{h} \leftarrow \mathtt{h}\cdot\mathtt{(1+}\gamma\mathtt{)} + \beta$) is injected at two layers. Because the backbone's embeddings and LM head are untouched, cross-entropy starts at pretrained quality; there is nothing to "recover."

Two consumer artifacts ship from this stage: `srt-adapter-v1.0` (semantic embeddings; the research series behind it culminated in a model-soup checkpoint, mean MTEB-STS 0.3744) and `zooL4nD3r-v0.1` (961 discourse communities). A separate diagnostic established the substrate's raw capacity for one downstream task: frozen-backbone features plus gradient boosting reach TruthfulQA-MC2 AUC 0.8656 $\pm$ 0.011 on Qwen2.5-7B (0.8563 on Gemma-2-2B, 0.8475 on Llama-3.2-3B), at the top of the published hidden-state-detector band (SAPLMA $\approx$ 0.72; INSIDE $\approx$ 0.78–0.85; EigenScore $\approx$ 0.80–0.85).

2.2 The Activation Verbalizer (Stage 4)

A ~12.7M-parameter head trained so that, given a mid-layer hidden vector `v` from the frozen backbone, it generates text whose own re-encoded activation matches `v`. Stage 4's lasting contributions are the measurement frame (§3) and the decoding result: with oracle best-of-K scoring (legitimate at deploy time, since `v` is by construction available), the verbalizer saturates the paraphrase ceiling at K=64 on Qwen2.5-7B. Its honest limits are equally important and appear in §9.

2.3 The linear cross-modal head (the invariance arc)

The simplest instrument in the program and the one that carries the headline: two linear maps (5376$\rightarrow$1024 each, ~22MB in bf16) trained with a symmetric InfoNCE objective on frozen-backbone states of paired images and captions, plus two per-modality centering vectors (~42KB). Nothing else. §6 and §7 show what it does and what it survives.

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3. Measurement discipline

The program's numbers are trustworthy because of four standing rules, each learned the hard way.

Rule 1: always correct for anisotropy. Transformer hidden states share a dominant direction; unrelated Qwen2.5-7B L20 states already have cosine $\approx$ 0.24 ($\|\mu\| \approx 55$), and raw cosine metrics are therefore uninterpretable. Every similarity in this program is computed after subtracting the relevant pool mean, and the raw anisotropy is reported alongside. In the visual channel the same rule reappears with a refinement (§6.3): the anchor population must be drawn from the query's own domain, and domain match beats population size.

Rule 2: anchor every scale. Stage 4 metrics are reported as ρ_norm $\in$ [0,1], normalized between a measured random-text floor (centered fve 0.510 on Qwen L20) and a measured same-source paraphrase ceiling (0.799), with replay (0.968) and nearest-neighbour retrieval (0.714) as reference points. A number without an anchored frame is not a result.

Rule 3: every ladder carries controls. Shuffled-pairs fits (capacity floors), train-size curves (memorization tests), held-out splits that are never touched by model selection, and leakage audits. The Karpathy benchmark run (§6.2) dropped 8,799 training pairs whose images belong to the Karpathy test and validation splits but had been moved into train2017 by COCO's 2017 re-partition, and retrained the head before evaluating.

Rule 4: respect encoding conventions. BOS-sensitivity (a bare re-encode drops gemma-4 replay from 0.9986 to 0.615), EOS-inclusive scoring matched between evaluation and calibration, and per-modality centering are all mandatory and all documented in the released code.

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4. What the taps read on a single substrate

Before invariance can mean anything, the readings themselves must be established on one substrate. On frozen Qwen2.5-7B the program established, with full details in the companion manuscripts:

- **Semantic structure**: the adapter's discourse/embedding vector supports competitive semantic similarity (the v1.0 product line and its research series).
- **Reflexivity and regime**: per-token $\hat{r}$ and a two-class regime signal, calibrated well enough that downstream ports report expected-calibration errors in the fourth decimal place (§5).
- **Recoverability of the state itself**: a single L20 hidden state is recoverable as text up to the paraphrase ceiling of the backbone under best-of-64 oracle decoding (ρ_cen $\approx$ 0.99 against the anchored frame), with zero-training nearest-neighbour retrieval ($\rho \approx 0.71$) as the deploy-time fallback that needs no generation at all.
- **Cross-architecture replication of the substrate claim**: the Stage-3 read-off survives 1-NN probes on Qwen3-8B and Mistral-7B-v0.3, and the NLA pipeline replicated on Llama-3.2-3B and Gemma-2-2B.

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5. Invariance axis one: architecture and scale, upward

The read-only adapter ports cleanly to radically different hosts, with only the ~16M side-channel heads trained and the backbone verified bit-exact against its reference forward pass in each case.

host	architecture	regime ECE	regime AUROC	community NMI
Qwen3-235B-A22B-FP8	94-layer MoE, 22B active	0.0005	0.986	0.62
gpt-oss-20b (MXFP4)	sliding-window MoE	0.0009	0.974	0.42

The 235B port required sharding across eight GPUs and a sliding-window-mask implementation verified byte-identical to the reference; the signals survived unchanged. Reflexivity correlation on gpt-oss-20b: Pearson 0.689. These ports established upward scale- and architecture-generality well before the question of *downward* invariance (§7) was posed.

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6. Invariance axis two: modality

6.1 The zero-shot discovery (SRT-Sunstone)

A 12.3M community read-out trained on **text only**, attached to frozen multimodal gemma-4-31B-it, interprets **images** with zero image training: CIFAR-10 image-to-word retrieval@1 of 0.93 against a chance rate of 0.10, and open-vocabulary retrieval of full sentences from a pool of ten thousand COCO captions it was never paired with. Cross-modal alignment peaks at roughly 80% of backbone depth and collapses at the final layer, replicating the late-is-surface signature seen on gpt-oss-20b.

The boundary study made the claim precise. On a random-dot autostereogram, whose figure exists only in binocular disparity, the read-out honestly reports texture; a simulated binocular-fusion front-end recovers the figure from the same pixels, and the read-out then names it, matching the true-silhouette profile. The capability gap was in the sensor, not the semiotics. A second boundary, the apparent inability to retrieve captions for synthetic graphics, was later shown to be substantially a *calibration artifact*: re-anchoring the image-side mean on a large natural-photo population moved the white-heart control's exact caption from rank 352 to rank 5 out of roughly ten thousand (§6.3).

6.2 The modality gap is anisotropic-linear

The question the discovery begged: what transformation connects the image states and the text states? We answered it with a controlled ladder on COCO pairs (gemma-4-31B-it, layer 47; evaluation split of 1,000 held-out val2017 images against their 5,000 captions, untouched by any model selection; shuffled-pairs control at every rung).

method	i2t R@1	R@5	R@10	t2i R@1
centered cosine (zero training)	0.288	0.523	0.648	0.173
orthogonal Procrustes (best variant)	0.247	0.506	0.656	0.164
trained linear, 117k pairs, InfoNCE	**0.661**	**0.911**	**0.967**	**0.506**
two-layer MLP, 117k pairs	0.567	0.887	0.943	0.448
shuffled-pairs control	0.001	0.008	0.013	0.002

Three findings. First, **a rotation cannot express the gap**: every orthogonal-Procrustes variant scores below doing nothing beyond centering, because the two modality means are nearly parallel

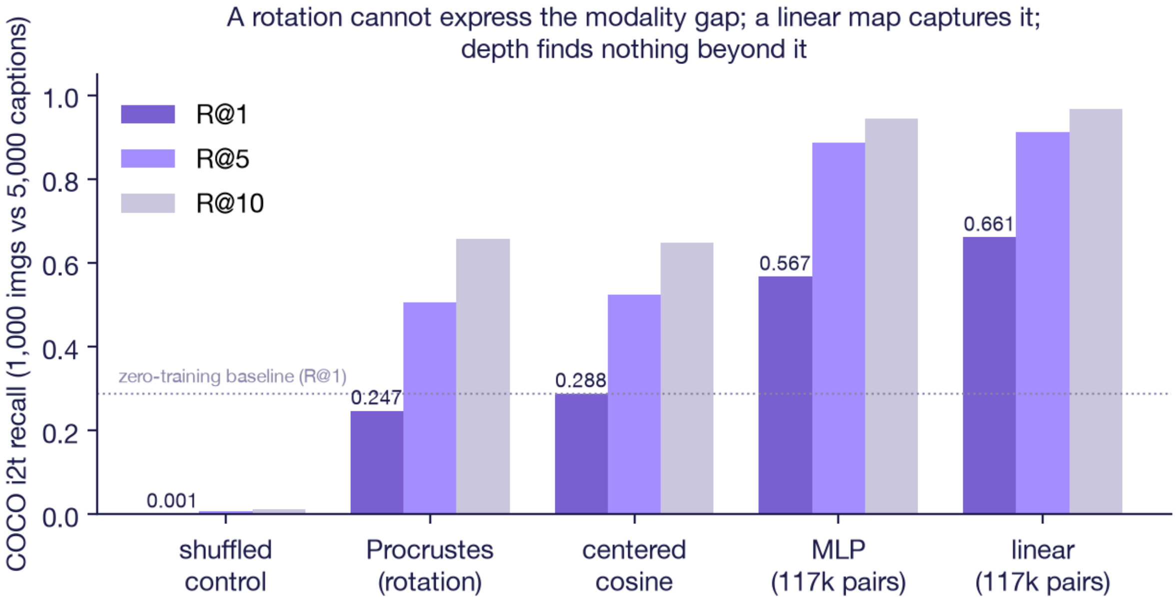

Figure 1: **Figure 1. The method ladder.** A rotation (orthogonal Procrustes) scores below doing nothing beyond per-modality centering; a single trained linear map captures the gap; a two-layer MLP with the same data never reaches it. Shuffled-pairs control at the floor.

(cos(μ_img, μ_txt) = 0.979) and the residual difference is a scale-and-shear that a norm-preserving map cannot fix. Procrustes' least-squares objective raises absolute paired similarity while destroying rank discrimination; the objective must optimize ranking. Second, **a single linear map captures the gap**, and its ceiling is still rising at the full COCO supervision (gains decelerate from roughly +0.10 to +0.06 R@1 per doubling; 0.661 is a lower bound). Third, **there is nothing beyond linear to find**: the MLP trails the linear map at every training size, and the gap *widens* with data (0.047 at 39k pairs to 0.094 at 117k). Thirty-three times more data gives the nonlinear model every chance to reveal structure a linear map cannot express; it instead falls further behind.

On the literature-standard **Karpathy 5k test**, leakage-controlled as described in §3, the head scores i2t R@1/R@5/R@10 = 0.416 / 0.710 / 0.818 (median rank 2) and t2i 0.292 / 0.567 / 0.685. This matches fully-trained 2018 dual encoders essentially digit for digit (VSE++: 0.413 / 0.711 / 0.812) from a linear map over a frozen chat model, using roughly three thousand times less pair data than CLIP-class systems. The claim is not "beats CLIP" (CLIP-class zero-shot sits near 0.58 at R@1); the claim is **no new model**: retrieval as a free rider on a host the deployment already runs.

6.3 The anchor rule, refined

The white-heart revision above generalizes Rule 1 to the visual channel with a twist a control forced on us: re-scoring the demo gallery's CIFAR-style thumbnails with the 4,000-photo COCO mean *degraded* their retrievals, while the same mean rescued the synthetic probes queried against a COCO pool. The anchor population must match the query's domain, and when domain and size conflict, domain wins: 150 in-domain images beat 4,000 out-of-domain ones. Practically this makes anchoring a calibration step, not a constant.

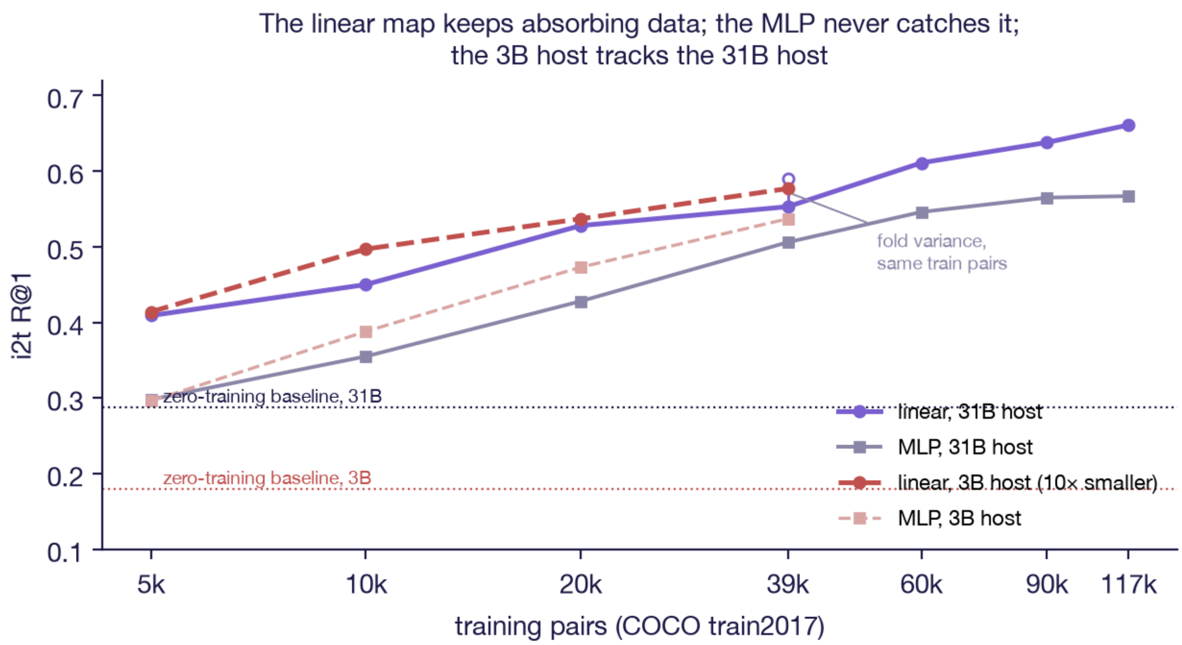

Figure 2: **Figure 2. Data scaling and the scale floor in one frame.** The linear map absorbs data faster than the MLP at every size on both hosts, and the 3B host's curve tracks the 31B host's: a ten-fold reduction in host scale costs nothing at matched training budget. The two 39k markers on the 31B linear curve show early-stopping fold variance on identical training pairs.

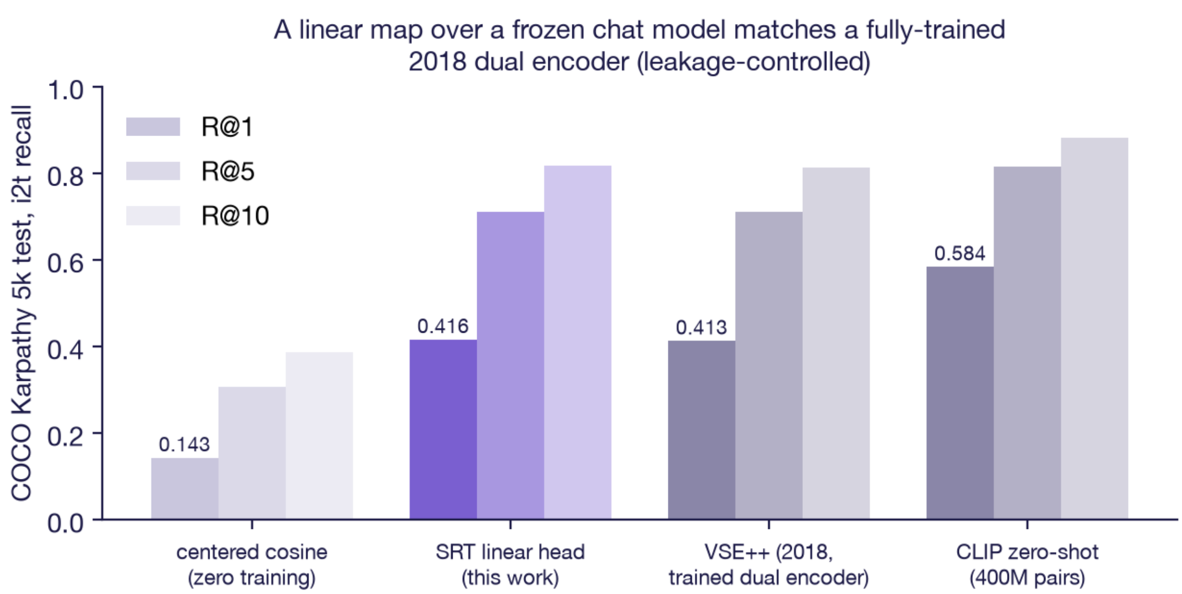

Figure 3: **Figure 3. Karpathy 5k placement (leakage-controlled).** The linear head over frozen gemma-4-31B-it matches VSE++, a fully-trained 2018 dual encoder, essentially digit for digit, and sits below CLIP-class zero-shot, which trained 400M parameters on 400M pairs.

6.4 What the head keeps is chosen by the objective

Compositionality benchmarks probe whether a retrieval system distinguishes "a horse eating grass" from "grass eating a horse." CLIP-class dual encoders famously sit near chance on the word-order splits of SugarCrepe. The shipped head inherits the same failure (swap splits at 0.50 to 0.57), which is at first surprising, because a diagnostic on the raw layer-47 states shows the substrate separates word-order swaps *more* sharply than object replacements (mean positive-negative cosine 0.396 vs 0.582). The information survives to the tap; the head discards it, because InfoNCE with random in-batch negatives never charges for discarding syntax. The failure belongs to the objective, not the encoder.

Fixing the objective recovers most of it. Adding rule-based compositional perturbations of each training caption as explicit hard negatives (noun swaps, then vocabulary replaces, then spatial preposition replaces and dependency-parsed adjective transfers, up to K=7 per caption, full-pool in the InfoNCE denominator) moves SugarCrepe macro accuracy from 0.631 to 0.705 across four retraining rounds, with gains landing mostly on the trained axes (relation negatives moved replace_rel +4.7 and attribute negatives moved add_att +5.1, neither reachable by reweighting the previous mix; the swap_att gain, by contrast, was 95% available from the weight dial alone, so per-axis attribution requires the dial as a control). A weight sweep gives the trade a measured frontier: from retrieval-first (clean i2t 0.661, macro 0.642) to compositionality-first (clean 0.600, macro 0.705), with intermediate weights dominated on both axes by w=1.0.

The arc ends at a wall that is itself a finding, twice over. The union of both negative families does not compose: per split it lands at approximately max of the parents, not the sum of their exclusive gains (macro 0.705 vs 0.703/0.685). Because the union run raised the per-batch negative count 1.75x alongside the mix change, mix and pressure were deconfounded with a pressure-matched control: the union subsampled to K=4 (same pool size as either parent, both families present) lands at macro 0.689, below the specialized parent. Mixing at fixed budget dilutes (replace_rel 0.758 to 0.735, add_att 0.695 to 0.659), and the union's small edge over the specialized parent was pressure buying back that dilution, not families composing. The two levers separate cleanly: the mix chooses which axes move, and the count pays for coverage, with the count lever cheaper in clean retrieval per unit of macro (the pressure-matched union also keeps clean i2t at 0.628 versus the full union's 0.600). The natural reading of the wall, that the 1,024 projection dimensions are a capacity budget, was then tested directly and refuted: retraining at proj_dim 2,048 and 4,096 with the same union negatives reproduces macro 0.705 exactly (0.698/0.705), so quadrupling the rank of the bilinear similarity buys nothing. The competition among trained properties (drift-nulling versus compositional margins versus clean retrieval) is a property of the objective and the data, not of width: a caption and its perturbation share one image-side target, and no projection of any rank can separate what the pooled image state does not distinguish. The last candidate, pooling itself, was then tested and also eliminated. A diagnostic on raw states was encouraging: scoring SugarCrepe by the maximum over four contiguous image-token bands instead of the global mean lifted exactly the order-sensitive splits (swap_obj +6.5 points from chance, swap_att +3.2, replace_rel +2.3), suggesting the mean was destroying real spatial signal. But the trained version of that hypothesis, a five-slot multi-vector head (four token bands plus the global mean per image, per-slot centering, max-over-slots InfoNCE, the same union negatives, re-encoding all 118,287 training images) lands at macro 0.702, inside the 0.703-0.705 band. Whatever spatial signal max-over-bands exposes on raw states, the trained linear readout already extracts an equivalent amount from the pooled state. A fifth lever closed the within-pair elimination: re-encoding the decomposition as 2x2 spatial quadrants (calibration first established that the backbone's image tokens form a row-major, aspect-matched grid of variable dimensions) reproduces swap_obj at 0.645 to the third digit, and retraining on the swap-family negatives alone regresses (macro 0.682, swap_obj 0.633): concentrating the objective on the permutation axis extracts nothing further.

External review then exposed two flaws in the aggregate itself, and repairing them reversed the attribution. First, the two add splits are 99% solvable by caption length alone (the negative is longer by construction; SugarCrepe's adversarial refinement controlled plausibility and fluency, not length), so the seven-split macro blends a text-side artifact into the measurement. Restated over the five length-clean splits the band is unchanged (0.69-0.71), and the trained heads score 0.66-0.71 on the add splits rather than 0.99, confirming that InfoNCE deletes length as a nuisance feature rather than exploiting it. Second, and decisively: every head in the elimination read gemma-4 states on both sides, so nothing in it could separate an image-side limit from a text-side one. A 2x2 tower factorial with Qwen2.5-VL-3B (layer 29, d=2048) as the second backbone, identical recipe in every cell (117,787 pairs, the same negative families re-encoded in whichever space the text tower lives in), settles the question. Clean five-split macro: gemma/gemma 0.704, gemma-image with qwen-text 0.665, qwen-image with gemma-text 0.690, qwen/qwen 0.661. Swapping the text tower costs 2.9 to 3.9 points; swapping the image tower costs 0.4 to 1.4. The band the elimination attributed to the image representation is set roughly 3.5x more strongly by the text representation, and a 10x smaller image tower read through gemma's text tower loses only 1.4 points. Continued review then imposed a second discipline on the per-split numbers: a blind word-order prior (bigram log-counts over the benchmark's own 4,345 unique positive captions, leave-one-out by caption text) scores 0.660 on the clean five splits with no image, no head, and no encoder, so every split is reported here as margin over that prior. The margin view sets an absolute floor and only that: the prior is blind to both towers, so it takes one value per split in every cell, and tower comparisons live in the raw table unchanged. Against the floor, the picture is: the gemma-text cells clear it (+0.044 and +0.031 on the clean five) while the qwen-text cells sit on it (+0.006 and +0.001), and the baseline itself carries error that point margins conceal. Two independent rebuilds of the prior agree within ±0.02 per split, which on swap_obj (n=245, prior CI ±0.062) means every cell's margin interval straddles zero against both rebuilds (`artifacts/nla/q4/prior_comparison.json`). The control that does adjudicate per cell, due to the same review, is mean-image ablation: the same head and texts with every image replaced by the split's mean image vector. On the qwen-text cells it is decisive in an unexpected direction: the ablation outscores the real images on the clean-five macro (0.693 vs 0.665 mixed; 0.687 vs 0.661 matched), with real images adding roughly +15 points on replace_obj and subtracting roughly 11 on the swap splits, where the trained text projection's caption prior alone scores 0.71+. Image content contributes exactly where object identity is at stake and degrades the text prior everywhere subtler (`artifacts/nla/q4/mean_image_control.json`; gemma-text cells pending re-encode). Two per-split observations from the raw table must be restated accordingly. swap_att, which follows the text tower in raw accuracy (0.69 and 0.66 under gemma text against 0.61 under qwen text), is itself largely solvable by the blind prior (0.67-0.69), so the raw effect is consistent with either better alignment or the bigger text model's stronger word-order prior showing through, and we no longer read it as a tower fingerprint. And swap_obj, which moves in no cell (0.600, 0.604, 0.620, 0.604), is underpowered at n=245 rather than demonstrably invariant: one cell's 95% interval is about ±6 points and the smallest resolvable cell difference (8.6 points) exceeds the largest tower effect seen anywhere in the table, so an effect of realistic size would be invisible there. Its margin over blind is directionally positive in all four cells but the interval straddles zero in every cell against both rebuilds of the prior; the split neither confirms nor rules out an image contribution at this n.

The correct summary of the wall is therefore not an image-side ceiling but a division of labor that is uniform across towers: the linear readout recovers the scene's inventory nearly in full and its arrangement hardly at all. Three direct measurements support that characterization. Scoring 80 COCO category prompts against 1,560 annotated images through the head gives mean per-category detection AUC 0.883 and per-image R-precision 0.543 against a 0.038 chance floor, so most object types present in a scene are recoverable from the single 1,024-dimensional vector. The worst-ranked of each image's five reference captions still lands at median rank 44 of 5,000, so the vector answers to every description of the scene rather than to a dominant subject. And retrieval is flat in scene complexity (r@5 0.906 at one annotated category, 0.887 at six or more). A 22 MB linear head on a frozen chat model closes 75% of the gap to CLIP ViT-B/32 on SugarCrepe by objective repair alone; the remaining gap is a property of the frozen pair, dominated by the text side, with an arrangement floor that no tested combination of towers, objectives, or decompositions moves (`artifacts/nla/q4/sugarcrepe_*.json`, `artifacts/nla/q4/w05_verdict_20260806.json`, `artifacts/nla/q4/width_null_20260807.json`, `artifacts/nla/q4/slot_pool_verdict_20260807.json`, `artifacts/nla/q4/sugarcrepe_mixed_v6.json`, `artifacts/nla/q4/sugarcrepe_qwen3b_v6.json`, `artifacts/nla/q4/sugarcrepe_cell4_v6.json`, `artifacts/nla/q4/inventory_A_multilabel.json`, `artifacts/nla/q4/blind_bigram_prior.json`, `artifacts/nla/q4/margin_over_blind.json`).

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7. Invariance axis three: scale downward, precision, and hardware

The final arc varied the three parameters a deployment actually controls and measured what the head noticed.

7.1 Host scale: 31B → 3B, no loss

The full ladder was re-run, identical code paths, on frozen Qwen2.5-VL-3B-Instruct (layer 29 of 36, d = 2048, the same 80%-depth heuristic). Every element of the fingerprint reproduced: the zero-training baseline works (R@1 0.180, 180× chance); Procrustes hurts (0.147); the trained linear map dominates (0.577 R@1, 0.870 R@5, 0.955 R@10 at 39k pairs, median rank 1); the MLP trails at every size; the shuffled control sits at 0.000. The decisive comparison is at matched training budget: the 3B host's 0.577 against the 31B host's 0.553–0.590 (the range reflecting early-stopping fold variance). Within noise, **a ten-fold reduction in host size costs nothing.** The linearly readable cross-modal correspondence is not an emergent luxury of scale; it is generic to the model class down to at least 3B, and because the retrieval read-out needs a single prefill pass rather than generation, 3B at 4-bit is Raspberry-Pi-class hardware.

7.2 Weight precision: bf16 → 4-bit, −0.011 R@1

The 3B head, trained on bf16 states, was applied **unchanged** to states from the same model loaded in 4-bit NF4:

configuration	i2t R@1	R@5	R@10
head on bf16 states (reference)	0.577	0.870	0.955
same head, unchanged, on 4-bit states	0.566	0.857	0.941
plus 42KB mean recalibration	0.569	0.868	0.948

The zero-training baselines are statistically identical across precisions (0.185 vs 0.180): four-bit quantization barely perturbs the representation geometry the head reads. A head retrained natively on quantized states lands on the same data-scaling curve as bf16 training, confirming that quantized states are fully usable substrate; but the cross-application result shows retraining is unnecessary. Data budget matters; precision does not.

7.3 Hardware and runtime: datacenter CUDA → consumer Apple Silicon

Finally, the original 31B head (trained on CUDA bf16 states in a datacenter) was tested against states produced by a different quantization of the same model (MLX 4-bit), running under different kernels on a consumer machine (Apple M2 Ultra). The protocol is cross-runtime self-retrieval over the full 5,000-caption calibration pool, both directions, six arms including two subspace controls, with the reference re-encoded fresh on CUDA to establish a reproducibility ceiling (`artifacts/nla/q4/head_space_validation_v2_20260805.json`).

The ceiling first: re-encoding the identical captions on the identical CUDA runtime reproduces the stored reference at 99.96% R@1, once duplicate captions are scored correctly (17 of the 5,000 rows are string-identical to another row, and returning an identical twin is not an error). Kernel nondeterminism alone therefore costs about 0.04%, and no cross-runtime number should be read against 100%. Across the runtime boundary, raw states retrieve their datacenter twins at 93.3% R@1 as-is. Mean-centering alone is inert (93.4%). **Through the head**, agreement rises to 96.8%, and to **97.0% with the 42KB per-runtime mean recalibration**.

Two control arms pin down where the text-side drift lives. Projecting both sides onto the top-1024 PCA basis of the reference, the high-variance subspace at the head's own output dimension, lands at 92.2%, below raw. A random orthonormal 1024-dimensional projection lands at raw (93.2%). Keeping the top tenth of the variance keeps the damage; the head's row space, which lives in the low-variance tail, is selectively clear of it. The drift is not a mean shift and not isotropic: it is concentrated in the high-variance subspace that the head's 5,376-to-1,024 projection discards.

A correction is owed here. An earlier draft of this section reported 98.4%/100% and attributed them to head space; a reader's code review established that the validation script never applied the head (it measured mean-centered raw states at a smaller pool, where the task saturates). The numbers above are from the corrected protocol, with the head applied, duplicate-aware scoring, and the subspace controls the same reader's null experiment called for.

The image side of the same boundary closes the remaining scope, and a further reader observation sharpened how to read it: the agreement metric transforms both sides identically, so a shared-mean frame error cancels there, while an end task scores queries against an external gallery frame and cannot hide it. The end tasks are therefore the instrument of record. The 1,000 evaluation images were encoded on both runtimes (mean of the layer-47 states over image-token positions). On image-to-text retrieval against the 5,000-caption gallery, the consumer runtime reaches **R@1 0.640 / R@5 0.903** against the datacenter reference's 0.670 / 0.919, through the unchanged head plus the 42KB recalibration; with the shipped training mean instead of the local one, R@1 drops to 0.401, so the image-side mean correction is worth 24 points. In the reverse direction, text-to-image with the same external gallery frame, the consumer runtime reaches **R@1 0.444 / R@5 0.740** against the reference's 0.503 / 0.808; the per-runtime mean buys +2.6 points there (and is neutral in a same-runtime control), with a roughly 6-point structured residual that no mean correction recovers, consistent with the subspace-control finding above. Both sides of the boundary carry a mean-frame component that the 42KB recalibration fixes; the text side additionally carries the high-variance structured drift, and pooled image states carry almost none. The claim's "at most a 42KB recalibration" clause does real work on both sides, and the honest tier summary is end-task: i2t at 95% of the datacenter reference, t2i at 88% (`artifacts/nla/q4/{image_head_space_validation,t2i_external_frame}_20260805.json`).

A final refinement, again reader-supplied, fixes the units for all of this. A constant displacement competes with the query it is added to, so the meaningful knob is not its raw norm but rho, the displacement's head-space gain divided by the median head-space query norm. In those units the two branches respond essentially identically to matched displacement, and the apparent image/text asymmetry in end-task damage is explained by the query-norm gap alone: the same cross-runtime mean delta (raw norms 22.7 and 21.5) lands at rho 1.33 on the image branch and 0.35 on the text branch. There is one displacement, arriving unequally, not two drift geometries. The same instrument also characterized a jitter-augmented retraining of the head: training against random-direction displacements bought 3.0x rho headroom against random directions but only 1.47x against the real, structured direction, which is measured evidence that augmentation priors for runtime drift should be drawn from measured drift families, not isotropic noise (`artifacts/nla/q4/round5_v2_direction_20260806.json`).

Acting on that evidence closes the loop. A head retrained with the measured drift family itself, each training sample perturbed by a real per-vector cross-runtime residual (3,000 measured MLX-minus-CUDA pairs, scaled U(0, 1.5)), internalizes the drift outright. On held-out data whose residuals were never seen in training: image-to-text with **no calibration at all reaches R@1 0.636**, matching what the original head needed the recalibration to achieve (0.634); text-to-image reaches 0.469 against the original head's 0.424 ceiling, the first crack in the structured residual after post-hoc affine correction and isotropic jitter both failed; the cross-runtime gap narrows to 0.2 R@1 points on image-to-text; and there is no measurable clean-performance tax (leakage-controlled same-runtime reference 0.660 vs the original head's 0.668, inside noise at this sample size). The residual is learnable structure that generalizes across inputs. On the drift-trained head the 42KB recalibration adds +2.2 i2t and nothing on t2i, against +24.4 and +2.5 on the original head: the recalibration and drift-trained retraining are substitutes for the mean component, not steps of a ladder, and the choice between them is a data-availability question (200 unpaired states versus 3,000 paired encodes from the target runtime) (`artifacts/nla/q4/v3_drift_head_eval_20260806.json`; `sunstone_linear_head_v3_drift.pt` is the shipped artifact and now serves the public demo).

An engineering note with strategic weight: on this consumer runtime the "state tap," the one piece of edge engineering the program had scoped as remaining work, turned out to require no work at all (the runtime exposes intermediate states natively). The full stack, a 17GB quantized 31B host plus the 44MB head, runs on a home computer with no GPU server involved.

7.4 Across model families: the cross-vendor bridge

The factorial of §6.4 also yields the program's strongest cross-family result, and it deserves its own axis. The mixed cells pair towers from different organizations, trained with no shared lineage, data, or weights: Google's gemma-4-31B on one side and Alibaba's Qwen2.5-VL-3B on the other, connected by the same single linear map after per-modality centering. Retrieval through the bridge: gemma images searched by qwen text reaches i2t R@1 0.611 against the matched gemma pair's 0.661, and qwen images searched by gemma text reaches 0.600, with MLP and shuffled controls behaving as on every other rung. Two model families that categorically never distilled each other align through one 44MB artifact at a five-point cost. This extends the invariance claim across the last axis a deployment can vary, the vendor, and it carries a practical corollary: an image index built on one backbone survives migration to another for the cost of a linear calibration rather than a fleet-wide re-encode. It also supplies a measured caution for the current policy debate: representational similarity between independently trained models is the norm, not forensic evidence of copying (`artifacts/nla/q4/mlp_align_mixed_v6.json`, `artifacts/nla/q4/mlp_align_cell4_v6.json`).

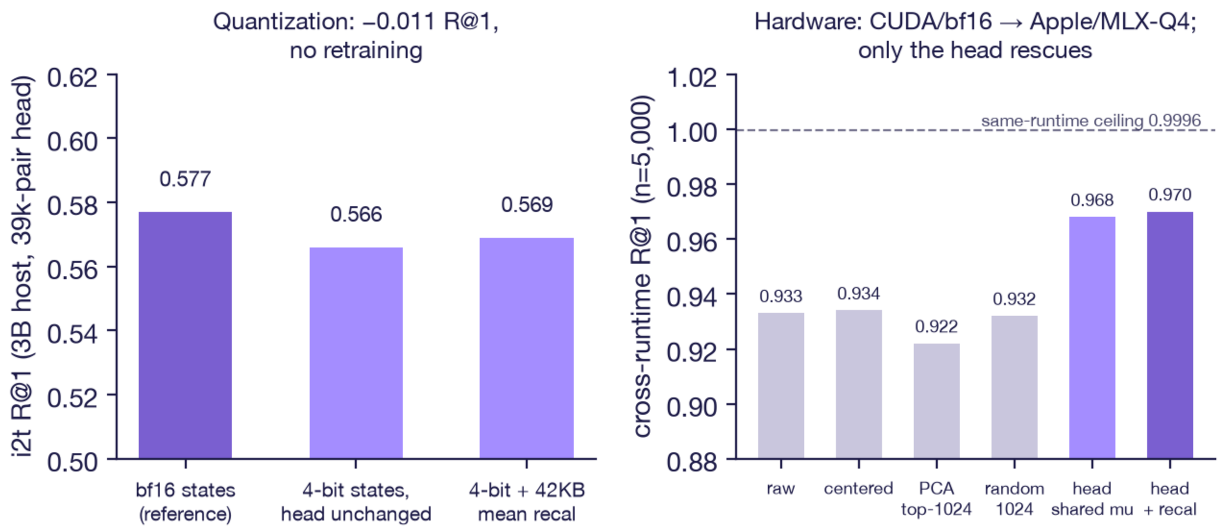

Figure 4: **Figure 4. Precision and hardware invariance.** Left: the bf16-trained head applied unchanged to 4-bit states loses 0.011 R@1; a 42KB mean recalibration recovers half. Right: across a simultaneous change of hardware, kernels, and quantizer, raw text states agree at 93.3% R@1 over the full 5,000 pool, centering and the subspace controls do not rescue, and the head lifts agreement to 97.0%, against a 99.96% same-runtime reproducibility ceiling.

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8. The invariance claim, assembled

axis	variation	result
architecture	dense 7B → sliding-window MoE 20B → 94-layer MoE 235B	read-out signals survive; calibration in the fourth decimal
modality	text → images, within one frozen host	shared space; gap is one linear map
host scale (down)	31B → 3B	identical fingerprint; no loss at matched data
weight precision	bf16 → 4-bit NF4	−0.011 R@1, head unchanged

axis	variation	result
hardware / runtime / quantizer	CUDA + bnb → Apple Silicon + MLX	text: 97.0% head-space agreement (n=5,000) vs a 99.96% ceiling, drift pinned to the high-variance subspace the head discards; end tasks on-device: i2t R@1 0.640 vs 0.670, t2i 0.444 vs 0.503, 42KB recal load-bearing on both sides
model family (vendor)	gemma-4 image tower × Qwen2.5-VL text tower, and the reverse	one linear map bridges unrelated families: i2t R@1 0.611 / 0.600 vs 0.661 matched

The unification: what the SRT instruments read is not a property of a checkpoint, a precision, a device, or a scale. It is a property of the **model class**, stable enough that one small artifact, trained once, reads it everywhere the class is instantiated. The deployment consequences follow immediately. A single trained head serves an edge device batch-tagging photographs overnight, a laptop doing interactive local search, and a datacenter serving a fleet; every tier runs the *same artifact* against the *same structure*; and because the artifact is linear, it can be audited by inspection, and because the backbone is frozen, the host's primary function carries zero regression risk. The tiers differ in latency and cost. They do not differ in capability.

We note the claim's shape: it is an *invariance* claim, and its strongest evidence is a set of results that would individually be reported as negatives. Rotation fails. Depth finds nothing. Data does not rescue the nonlinear model. Quantization does not matter. Scale does not matter. Each null removes a way the structure could have been fragile.

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9. The negative-results ledger

The program treats documented negatives as first-class findings; the headline could not be trusted without them.

1. **The greedy gap.** On every host, greedy decoding from the activation verbalizer collapses far below retrieval; on instruction-tuned hosts the best-of-K curve never reaches the retrieval baseline at any practical K (+0.017 per doubling on gpt-oss-20b and gemma-4-31B-it, against +0.10 on base Qwen). Retrieval, not generation, is the honest decode on tuned hosts.
2. **The base-versus-tuned conjecture, refuted by its own control.** An earlier draft conjectured that instruction tuning collapses the manifold sampling must traverse. A within-family control (identical pipeline on the gemma-4 base checkpoint) produced an indistinguishable K-curve (+0.013 vs +0.015 per doubling). Whatever sets verbalizability is in the pretrained substrate, not the alignment stage.
3. **Draft-conditioning, refuted with a mechanism.** Conditioning the verbalizer on the retrieval neighbour's text adds 0.03 nats of predictive value for the gold text; cross-entropy training therefore has no gradient toward using it.
4. **Causal injection nulls.** Forcing community prototypes into the decode path produces near-zero causal movement on v8a (every diagonal effect within ±0.002 nats; the pathway

verified live by a 100×-magnified probe moving CE by +2.27 nats), and the released adapters' community-conditioning acts as a uniform style bias rather than a per-sign disambiguator. The diagnostic half of the program is robust; the causal-steering half is scoped out.

5. **The sensor boundary.** The autostereogram result: no read-out can report structure the encoder cannot compute. Supply the missing computation (simulated binocular fusion) and the boundary moves.
6. **The anchor artifact.** A published boundary claim (synthetic graphics "outside the domain") was substantially a calibration error, corrected and re-scoped in §6.3. We keep the original claim visible in the record because the correction is itself a finding about measurement discipline.
7. **Negative families do not compose, and width does not rescue them.** The union of two hard-negative families that individually moved their own SugarCrepe axes lands at approximately the per-split maximum of its parents, and compositional training erases a drift-trained head's family-nulling. The obvious explanation, 1,024 dimensions as a capacity budget, was tested and refuted: projections of rank 2,048 and 4,096 reproduce macro 0.705 exactly. The competition among trained properties lives in the objective and the data, not the width (§6.4, §7.3).
8. **The wall was attributed to the wrong tower.** Five levers eliminated within a single backbone pair supported "the wall is the layer-47 image representation." A 2x2 tower factorial with a second backbone reversed the attribution: the text tower sets the band roughly 3.5x more strongly than the image tower, the seven-split macro carried a caption-length artifact on its two add splits, and a blind word-order prior then showed the swap_att "tower fingerprint" to be largely the text prior showing through, with the swap_obj "floor" underpowered at n=245 rather than invariant. An elimination inside one pair cannot attribute a limit to either side of the pair, and raw split accuracy cannot be attributed to alignment without a blind-prior margin; both corrections and both methods are due to external review (§6.4).

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10. Semiotic interpretation

The program began in Peircean semiotics, and the invariance result gives the theory its sharpest empirical footing to date. The *interpretant*, the effect a sign produces in an interpreting system, was hypothesized in Stages 1–2 to be a real, structured object in representational space rather than a theoretical convenience. The evidence now reads: the interpretant-bearing structure (i) exists in frozen substrates that were never trained to expose it, (ii) is shared across the modality in which a sign arrives (word, photograph, or decoded depth map complete the same interpretant, and name the recovered figure identically to the real one), (iii) is *linearly* situated, so that reading it never requires reconstructing the system that produced it, and (iv) is invariant to the physical realization of the substrate, which is exactly what one would demand of a structural property of meaning as opposed to an artifact of a particular network.

The program's law-like battery on contestedness (dissociation, coupling, causal forcing, and a diachronic test over 11,876 dated articles spanning 1770–1964, where contested political signs' measured divergence drifts above concrete controls with a difference-in-differences of +0.048, permutation p = 0.007, crossing over in the early twentieth century) locates the semiotic load in the sign's representation itself rather than in any injectable community signal, consistent with the causal nulls of §9.4.

11. Related work

The measurement frame descends from the anisotropy literature (Ethayarajh 2019). The cross-modal ladder engages the cross-lingual embedding alignment tradition (orthogonal maps between separately trained spaces) and finds its central tool inapplicable here, for a reason worth stating: separately trained embedding spaces differ by rotation-like transforms, but two modalities inside one jointly trained substrate differ by translation and anisotropic scale. The retrieval baselines are the standard COCO lineage (Karpathy & Fei-Fei 2015; VSE++, Faghri et al. 2018; CLIP, Radford et al. 2021). The hidden-state diagnostic band is SAPLMA (Azaria & Mitchell 2023), SAR (Duan et al. 2024), and INSIDE/EigenScore (Chen et al. 2024). The adapter's contrastive components use supervised contrastive learning (Khosla et al. 2020). The theoretical frame is Peirce, read through Kockelman's semiotic stance and Silverstein's metapragmatics (Lancaster 2025, 2026a).

12. Limitations

The invariance evidence spans 3B to 235B, three architectures, two quantization schemes, two hardware stacks, and one cross-vendor family pair (gemma-4 × Qwen2.5-VL, both directions); other family pairs are unmeasured, and the ceiling numbers are gemma-4's. The linear ceiling at full COCO supervision is a lower bound, not a measured plateau. Sub-3B hosts are untested. The 4-bit hardware result uses NF4 and MLX-Q4; other quantizers should track but are unmeasured. Retrieval quality inherits pool coverage, and anchoring is a genuine calibration step (§6.3) that deployments must perform. Finally, the causal (injection/steering) half of the original program remains an open negative: these are reading instruments.

13. Conclusion

The SRT program set out to show that frozen language models are instrumentable substrates. It ends with something stronger: the structure the instruments read is invariant across the entire deployment envelope, scale, precision, and silicon, and cross-modally linear within a host. The practical corollary is a new class of artifact, small enough to ship as a configuration file, honest enough to audit by inspection, that grants a capability wherever its model class runs. The theoretical corollary is that meaning, as these substrates organize it, is a stable, linearly readable, realization-independent structure. Train once, read everywhere.

14. Artifact index

Models (Hugging Face, `RiverRider/`): `srt-adapter-v1.0`, `srt-adapter-v8a`, `zooL4nD3r-v0.1`, `srt-adapter-qwen3-235b`, `srt-adapter-gptoss20b`, `Gemma-4-31B-it-SRT-Sunstone`, `srt-sunstone-linear-he` (the 22MB head + anchors), `srt-nla-av-v1`, `srt-nla-av-gemma4`, `srt-nla-av-llama32-3b`.

Data and results (`RiverRider/srt-nla-gemma4-artifacts`): `procrustes/` (ladder results, fitted maps, 118k pair encodings, Karpathy eval), `scalefloor/` (3B replication + head), `q4/` (quantization-drift results + native-Q4 head), `overnight/` (run logs and summaries).

Code (github.com/space-bacon/SRT): the adapter and NLA packages (`srt/`, `srt_introspect/`), the experiment scripts (`scripts/gemma4_procrustes_xmodal.py`, `gemma4_encode_pairs.py`, `gemma4_mlp_align.py`, `gemma4_karpathy_eval.py`, `q4_drift_eval.py`, `local_sunstone.py`), and the engineering guide (`docs/CROSSMODAL_LINEAR_HEAD.md`).

Demos: `RiverRider/srt-sunstone`, `RiverRider/srt-showcase`, `RiverRider/srt-nla-gptoss20b-trace`, `srt-adapter-v1.0-demo`.

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